

Asintoti - Esercizi

mercoledì 4 dicembre 2024 20:22

N°4

$$y = 2x^3 + x$$

$$D = \mathbb{R} \rightarrow (-\infty, +\infty)$$

↓
NON POSSO
AVERE ASINTOTI

$$\lim_{x \rightarrow -\infty} 2x^3 + x = -\infty - \infty = -\infty$$

$$\lim_{x \rightarrow +\infty} 2x^3 + x = +\infty + \infty = +\infty$$

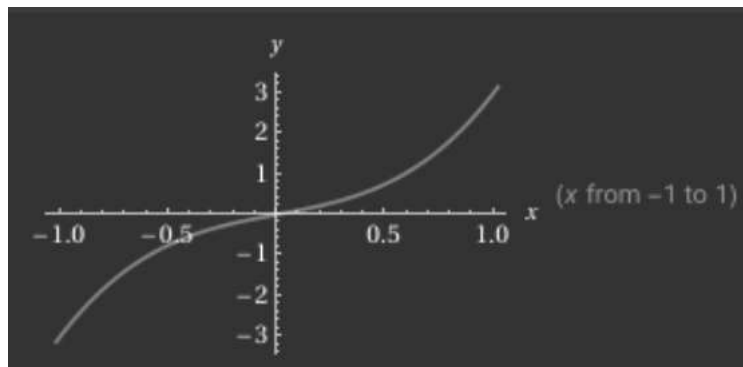
NON
ASI

$$m = \lim_{x \rightarrow +\infty} \frac{f(x)}{x}$$

$$m = \lim_{x \rightarrow \pm\infty} \frac{2x^3 + x}{x} = \lim_{x \rightarrow \pm\infty} \frac{x(2x^2 + 1)}{x} = +\infty$$

NON ABBIAMO ASINTOTO
OBLIQUO

INFATTI
NEL GRAFICO
NON CI SONO
ASINTOTI



N°5

$$y = \frac{x^2 - 4}{x^2 - 9}$$

1. DOMINIO

$$x^2 - 9 \neq 0 \rightarrow x^2 \neq 9 \rightarrow x \neq \pm 3$$

$$D = \mathbb{R} - \{-3, 3\}$$

$$D = (-\infty, -3) \cup (-3, 3) \cup (3, +\infty)$$

2. PARI/DISPARI

$$PARI (f(-x) = f(x))$$

$$\frac{(-x)^2 - 4}{(-x)^2 - 9} \stackrel{?}{=} \frac{x^2 - 4}{x^2 - 9} \rightarrow PARI$$

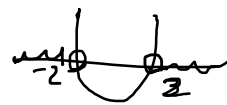
3. INTERSEZIONI

$$\begin{cases} \frac{x^2 - 4}{x^2 - 9} = y \\ y = 0 \end{cases} \rightarrow \begin{cases} \frac{x^2 - 4}{x^2 - 9} = 0 \\ y = 0 \end{cases} \rightarrow \begin{cases} x^2 = 4 \\ y = 0 \end{cases} \rightarrow \begin{cases} x = \pm 2 \\ y = 0 \end{cases}$$

$$\begin{cases} \frac{x^2 - 4}{x^2 - 9} = y \\ x = 0 \end{cases} \rightarrow \begin{cases} y = \frac{-4}{-9} \\ x = 0 \end{cases} \rightarrow \begin{cases} y = \frac{4}{9} \\ x = 0 \end{cases} \quad (0, \frac{4}{9})$$

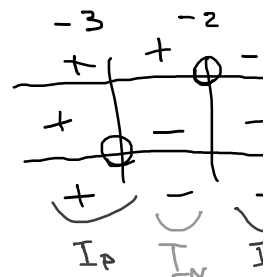
4. SEGNO

$$N \frac{x^2 - 4}{x^2 - 9} \geq 0$$



$$N x^2 - 4 > 0 \rightarrow x^2 > 4 \rightarrow x > \pm 2$$

$$D x^2 - 9 > 0 \rightarrow x^2 > 9 \rightarrow x > \pm 3$$



$$I_P = (-\infty, -3) \cup (-2, 2) \cup (3, +\infty)$$

$$I_N = (-3, -2) \cup (2, 3)$$

5. ASINTOTI

$$D = (-\infty, -3) \cup (-3, 3) \cup (3, +\infty)$$

$$\bullet \lim_{x \rightarrow \pm\infty} \frac{x^2-4}{x^2-9} = \lim_{x \rightarrow \pm\infty} \frac{x^2(1-\frac{4}{x^2})}{x^2(1-\frac{9}{x^2})} = 1 \quad \text{ASINTOTIA ORIZZ}$$

$$\bullet \lim_{x \rightarrow -3^-} \frac{x^2-4}{x^2-9} = \frac{9^+-4}{9^+-9} = \frac{5}{0^+} = +\infty$$

$$\bullet \lim_{x \rightarrow -3^+} \frac{x^2-4}{x^2-9} = \frac{9^- - 4}{9^- - 9} = \frac{5}{0^-} = -\infty$$

$$\bullet \lim_{x \rightarrow 3^-} \frac{x^2-4}{x^2-9} = \frac{9^- - 4}{9^- - 9} = \frac{5}{0^-} = -\infty$$

$$\bullet \lim_{x \rightarrow 3^+} \frac{x^2-4}{x^2-9} = \frac{9^+ - 4}{9^+ - 9} = \frac{5}{0^+} = +\infty$$

6. GRAFICO

